Euclid's Elements — Book I
Proposition 6

Formal Reconstruction — Technical Documentation

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Equal angles imply equal opposite sides

1 Introduction

Proposition I.6 is the converse of the isosceles-triangle theorem. Where Proposition I.5 shows that equal sides imply equal opposite angles, Proposition I.6 starts from

$$\angle ABC \cong \angle ACB$$

and concludes

$$AB \cong AC.$$

Euclid proves the result by reductio. Assuming that the two sides are unequal, he cuts from the longer one a segment equal to the shorter and uses SAS to force an angle to equal a proper part of itself. The current formal reconstruction is organized differently. The public theorem applies the already derived ASA side theorem to the same triangle with its two base vertices interchanged.

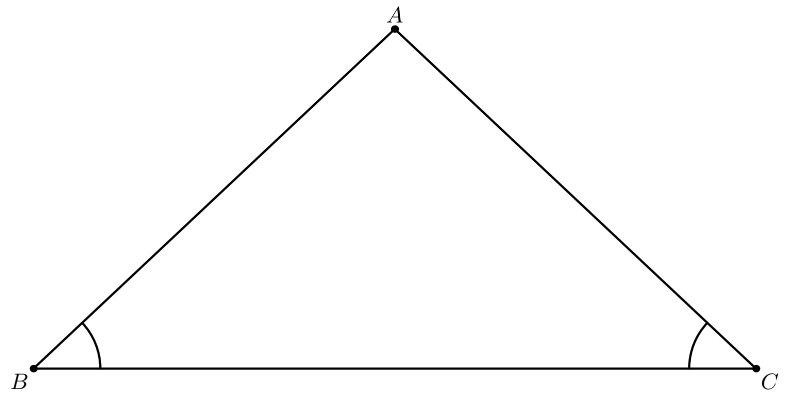
The present audit also descends one layer below the public call. The theorem `hilbert_asa_sides` is not a content-free wrapper: its proof contains an auxiliary segment construction, SAS, Hilbert III.4 angle-construction uniqueness, and an incidence-uniqueness argument. Those geometric steps are recorded explicitly below.

2 The Classical Configuration

Let ABC be a nondegenerate triangle with equal base angles:

$$\angle ABC \cong \angle ACB.$$

The desired conclusion is $AB \cong AC$.



$\angle ABC \cong \angle ACB$

Figure 1: The data of Proposition I.6: a nondegenerate triangle with $\angle ABC \cong \angle ACB$.

2.1 Euclid's reductio construction

Assume that the two sides are unequal. Interchanging B and C if necessary, suppose that AB is the longer side. Proposition I.3 cuts from AB a segment DB congruent to AC, with

$$A - D - B, \qquad DB \cong AC.$$

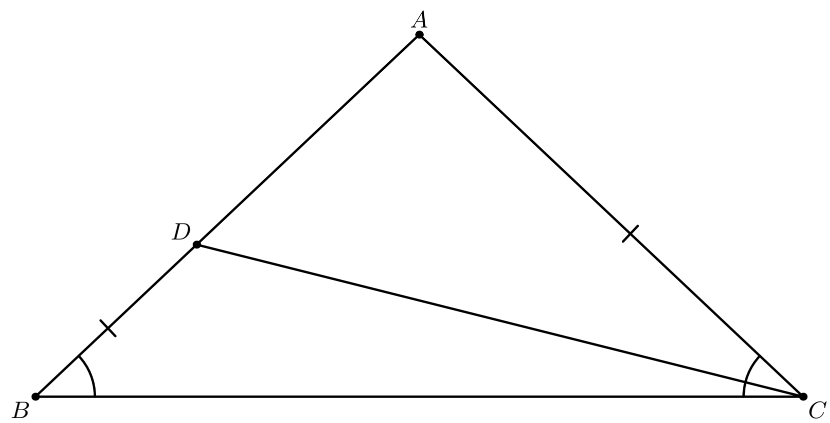
Join D to C.

Because D lies on AB, the angle DBC is the same angle as ABC. Hence the hypothesis gives

$$\angle DBC \cong \angle ACB.$$

Together with $DB \cong AC$ and the common side BC, Proposition I.4 applies to $\triangle DBC$ and $\triangle ACB$.

SAS on $\triangle DBC$ and $\triangle ACB$



$A - D - B, \quad DB \cong AC$

Figure 2: Euclid's contradiction configuration. Proposition I.3 gives $A - D - B$ and $DB \cong AC$; after joining D to C, SAS compares $\triangle DBC$ with $\triangle ACB$.

SAS gives, in particular,

$$\angle BCD \cong \angle ABC.$$

By the hypothesis the latter angle is congruent to $\angle ACB$. But the ray CD lies inside $\angle ACB$, so $\angle BCD$ is a proper part of $\angle ACB$. Thus the whole angle would be congruent to a proper part of itself, contradicting Common Notion 5. Therefore the unequal-side assumption is impossible.

The classical dependency chain is therefore

$$\text{I.3} + \text{I.4} + \text{Common Notion 5} \longrightarrow \text{I.6}.$$

No use of Proposition I.5 is required in the proof of its converse.

3 Statement

Theorem 1 (Euclid I.6)**.** *Let ABC be a nondegenerate triangle. If*

$$\angle ABC \cong \angle ACB,$$

then

$$AB \cong AC.$$

The current Lean declaration has exactly this shape:

```
theorem euclid_proposition_6
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not Collinear Geo A B C)
    (hAngles : Geo.AngleCongruent A B C A C B) :
    Geo.Congruent A B A C := by
  ...
```

Diagrammatic audit. The two classical figures are retained: the input triangle and the single stable auxiliary configuration after the point D has been cut from the longer side. No further classical snapshot is needed, because SAS and the proper-angle contradiction add no new point or incidence state.

The formal layer requires one additional two-panel figure. Panel (a) specializes the auxiliary construction hidden inside `hilbert_asa_sides`: when the theorem is applied to the two vertex orders BCA and CBA, it constructs a point X on the ray BA satisfying $BX \cong CA$. Panel (b) records the subsequent rigidity step: Hilbert III.4 forces the ray CX to coincide with the ray CA; since X also lies on the line BA, incidence uniqueness identifies X with A. The construction then yields $AB \cong AC$.

Thus the audited set is

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in } 3 \text{ figures}.$$

4 Proof Architecture of the Reconstruction

At the public Book I level the proof is short:

$$\begin{array}{c} \angle ABC \cong \angle ACB \\ \downarrow \\ \text{reorient the two equal angles and the common side } BC \\ \downarrow \\ \texttt{hilbert\_asa\_sides on } BCA \text{ and } CBA \\ \downarrow \\ AB \cong AC. \end{array}$$

The current source prepares two cyclic noncollinearity variants, reverses the common segment BC to CB, reorients the given angle congruence, and then calls

```
have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    C B A
    hBCA
    hCBA
    hBC
    hAngleB
    hAngleC
```

The final endpoint reversals are representational only.

The deeper mathematical engine is:

$$\begin{array}{c} \text{ASA data for } ABC \text{ and } A'B'C' \\ \downarrow \\ X \in \mathrm{ray}(B'C'), \quad B'X \cong BC \\ \downarrow \text{ SAS} \\ \angle B'A'X \cong \angle BAC \\ \downarrow \text{ given angle at } A \\ \angle B'A'X \cong \angle B'A'C' \\ \downarrow \text{ Hilbert III.4 uniqueness} \\ \mathrm{ray}(A'X) = \mathrm{ray}(A'C') \\ \downarrow \text{ incidence uniqueness} \\ X = C' \\ \downarrow \\ AC \cong A'C', \quad BC \cong B'C'. \end{array}$$

For I.6 this specializes to $X \in \mathrm{ray}(BA)$ with $BX \cong CA$, followed by $\mathrm{ray}(CX) = \mathrm{ray}(CA)$, and therefore $X = A$.

5 Hilbert Reconstruction

5.1 The public theorem

The formal Proposition I.6 contains no reductio and introduces no local geometric witness. It compares the triangle with itself after interchanging B and C. The side between the two equal

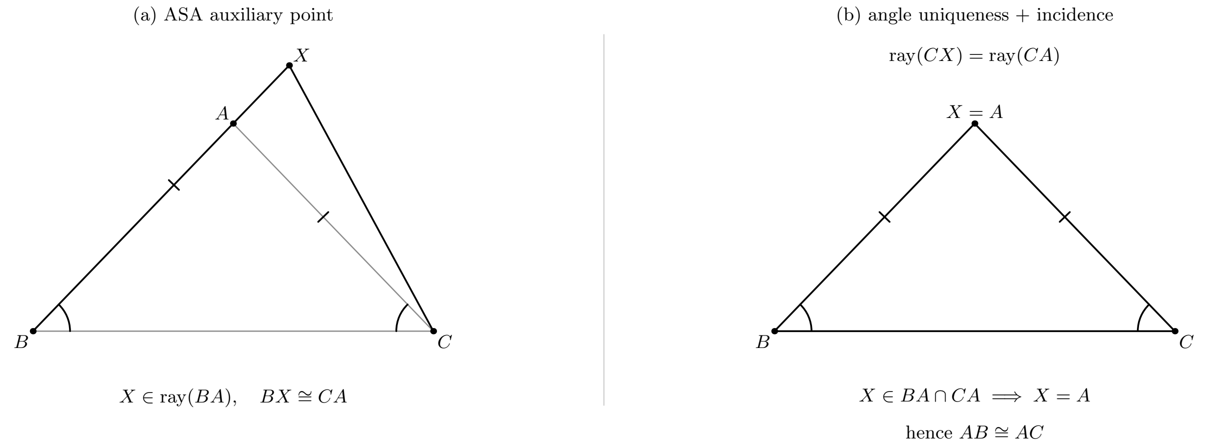


Figure 3: The formal ASA engine specialized to Proposition I.6. Left: the segment-construction witness X lies on the ray BA and satisfies $BX \cong CA$. Right: angle-construction uniqueness forces CX onto the ray CA; the two carrier lines meet at A, hence $X = A$ and $AB \cong AC$.

angles is the same segment BC, so ASA immediately supplies the two remaining side congruences. The decisive public dependency is therefore

$$\texttt{hilbert\_asa\_sides} \longrightarrow \texttt{euclid\_proposition\_6}.$$

5.2 The hidden ASA witness

The general theorem `hilbert_asa_sides` performs the geometric work that the public I.6 theorem does not show. It constructs X on the ray $B'C'$ so that $B'X \cong BC$. The construction deliberately puts X on the same side of the reference line as C'. SAS then transports the first given angle to the auxiliary triangle.

The second prescribed angle is now a uniqueness condition. Hilbert's angle-construction axiom III.4 says that on a fixed side of a fixed base ray there is only one ray making the prescribed congruent angle. Consequently $A'X$ and $A'C'$ are the same ray. Since X and C' already lie on the same carrier through B', the two distinct carriers would otherwise have two common points. Incidence uniqueness therefore gives $X = C'$.

This is the actual mathematical content hidden below the public ASA call. It uses only neutral incidence, order, and congruence structure; no Euclidean parallel axiom enters Proposition I.6.

6 Lean Formalization

The current production theorem first derives the cyclic noncollinearity variants

```
have hBCA : not Collinear Geo B C A := ...
have hCBA : not Collinear Geo C B A := ...
```

and prepares the common side in the required orientation:

```
have hBC : Geo.Congruent B C C B := by
  exact
    (Geo.congruent_reverse_second
```

```
    B C B C).mp
    (hilbert_congruent_reflexive Geo B C)
```

The hypothesis is then reversed to provide the two ASA angle inputs. After `hilbert_asa_sides` returns the side conclusion, the proof closes by endpoint reversal:

```
exact
  (Geo.congruent_reverse_second
    A B C A).mp
    ((Geo.congruent_reverse_first
      B A C A).mp hASA.1)
```

There is no proposition-local helper theorem, no case split, no segment trichotomy, and no segment layoff in `Proposition06.lean` itself. Those items appeared in an older documentation layer and are removed from the current formal dependency description.

7 Relationship with Earlier Propositions

7.1 Proposition I.3

I.3 is a genuine classical dependency. It creates the point D on the assumed longer side with $DB \cong AC$. The current Lean theorem does not call `euclid_proposition_3`; the corresponding construction has been absorbed into the lower ASA theorem in a different form.

7.2 Proposition I.4

I.4 is the decisive classical congruence step. SAS compares $\triangle DBC$ and $\triangle ACB$ and forces the angle equality that contradicts the proper-part relation. The formal ASA engine also uses a reusable SAS theorem internally, but it is not a proposition-level call to `euclid_proposition_4`.

7.3 Proposition I.5

I.6 is the converse of I.5, but I.5 is not a dependency of Euclid's proof of I.6. Together the two propositions establish

$$AB \cong AC \quad \iff \quad \angle ABC \cong \angle ACB$$

for a nondegenerate triangle.

7.4 Transition to Proposition I.7

The classical proof of I.7 uses I.5, not I.6. Thus I.6 is conceptually part of the early isosceles theory, but it should not be inserted as a spurious edge into the classical dependency DAG of I.7. The formal development, in turn, reorganizes both results around stronger reusable congruence and uniqueness theorems.

8 Hilbert Components Used

The actual formal dependency layer relevant to I.6 is:

- segment construction on a prescribed ray inside `hilbert_asa_sides`;
- same-ray and same-side transport;
- SAS angle and side consequences;
- Hilbert III.4 angle-construction uniqueness;
- line uniqueness from plane incidence;
- endpoint and angle orientation symmetries;
- the final ASA side theorem exposed to Proposition I.6.

Segment trichotomy, a two-branch comparison of AB and AC, and a Book-Zero layoff theorem are *not* dependencies of the current production proof.

9 Dependency Summary

The two architectures should be kept separate.

Classical.

$$\begin{gathered}\angle ABC \cong \angle ACB \\ \downarrow \\ \text{assume } AB \neq AC,\ \text{choose the longer side} \\ \downarrow \\ \text{I.3} : A - D - B,\ DB \cong AC \\ \downarrow \\ \text{I.4} : \triangle DBC \cong \triangle ACB \\ \downarrow \\ \angle BCD \cong \angle ACB \\ \downarrow \\ \text{whole angle equals a proper part} - \text{contradiction} \\ \downarrow \\ AB \cong AC.\end{gathered}$$

Formal.

$$\begin{gathered}\angle ABC \cong \angle ACB \\ \downarrow \\ \text{view one triangle as } BCA \text{ and } CBA \\ \downarrow \\ BC \cong CB \\ \downarrow \\ \texttt{hilbert\_asa\_sides} \\ \downarrow \\ AB \cong AC \\ \downarrow \\ \texttt{euclid\_proposition\_6}.\end{gathered}$$

Inside the reusable ASA theorem, the essential hidden geometric chain is

$$\text{segment construction} \to \text{SAS} \to \text{III.4 uniqueness} \to \text{incidence uniqueness.}$$

10 Audit Status

This documentation revision was checked against the current production `Proposition06.lean`, the current ASA theorem in the Hilbert axiom layer, and the preserved Book1R chapter. The Lean production file was not modified.

No fresh `lake build` and no new `#print axioms` run are claimed by this documentation audit. The PDF and all Asymptote figures in the deliverable were compiled and visually inspected.

11 Conclusion

Proposition I.6 is a particularly clean example of proof reorganization. Euclid's argument uses an explicit cut on the longer side and SAS inside a reductio. The formal Book I theorem instead recognizes the statement as the side conclusion of an already established ASA theorem.

The short public proof should not be mistaken for absence of geometry. The ASA theorem below it contains a genuine witness X, a segment construction, SAS, angle-construction uniqueness, and an incidence-rigidity step. Once that lower theorem is available, however, Proposition I.6 itself reduces to vertex reorientation and a single interface call.